Alyssa Rivera

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EDUCATION

San Diego State University | B.A. Applied Arts and Sciences, Computer Science May 2025

SKILLS SUMMARY

- Programming Languages: C++, C#, Python, JavaScript, SQL, HTML, Shell Scripting, HLSL, ShaderLab
- Frameworks & Tools: Unity, Flask, NumPy, pandas, Git, GitHub, PowerShell, WPF, PostgreSQL, SQLite, Visual Studio
- Testing: Integration testing, black-box testing, API testing, risk assessment

SOFTWARE ENGINEERING PROJECTS

Frozen Assets, 2024 (*Web-Based Application)*

Served as a Test Manager for a full-stack inventory and logistics management platform developed using Python, Flask, SQL, JavaScript, Vue.js, Bootstrap, and GitHub. The system integrated order entry, inventory management, shipment tracking, invoicing, and trouble-ticket management into a modular architecture supporting role-based access control, inventory synchronization, and transactional consistency. Led QA planning, integration testing, risk assessment, and JAD documentation efforts within a six person development team.

- Authored Joint Application (JAD) documentation by gathering system requirements, defining testing strategies, and documenting subsystem workflows, establishing a foundation for development, testing, and coordination.
- Designed and managed integration test plans by creating test cases, validation procedures, and QA documentation, ensuring reliable communication between order entry, inventory, shipment tracking, and ticket management subsystems.
- Validated authentication, authorization, and transaction processing workflows through integration, API, usability, and role-based security testing, improving system reliability and reducing defects.
- Conducted stress testing and concurrency using pytest, SQL validation queries, and logging tools, identifying synchronization issues that could lead to inaccurate inventory states during simultaneous transactions.
- Used pytest, SQL validation workflows, and logging/debugging tools to identify transaction failures, validation inconsistencies, and subsystem communication defects.
- Verified input validation logic and database transaction behavior by executing structured test scenarios across customer and administrator roles, reducing failed transactions and improving data consistency.
- Supported iterative releases by executing deployment validation and regression testing throughout development, helping the team identify defects earlier and improve overall system stability.

Tea Steeper Agent, 2025

Developed an embedded tea-steeping assistant in C++ on a Raspberry Pi using a DS18B20 temperature sensor to automate temperature monitoring and steep-time management. The system allowed users to select tea type and strength, dynamically calculating target temperatures and steep durations while continuously monitoring live sensor data until brewing conditions were met. Built using object-oriented design, state machine architecture, and Makefile build automation, the project emphasized hardware/software integration, sensor validation, and embedded systems development through iterative testing and debugging.

- Implemented hardware communication and DS18B20 sensor integration by developing temperature acquisition and threshold-detection logic, enabling real time monitoring of water temperature through the brewing process.
- Designed a state-machine architecture to manage tea selection, temperature monitoring, steep timing, and completion notifications, creating a reliable and maintainable workflow for user interactions.
- Validated sensor accuracy and runtime state transitions through integration testing and debugging, improving measurement reliability and ensuring correct execution of brewing sequences.
- Built automated Makefile compilation workflows to streamline dependency management and accelerate testing cycles.
- Resolved hardware/software integration challenges by troubleshooting sensor calibration, temperature propagation, and data-parsing issues, improving consistency between sensor readings and application behavior.
- Executed validation testing across multiple tea types and steep strength combinations to verify temperature calculations, timing logic, and overall system reliability.

[Cubelet, 2024]

Served as a level designer on a five person team developing a puzzle-horror game in Unity using C#, Rigidbody physics, ScriptableObjects, coroutines, Cinemachine, and the Unity Input System. The project combined physics-based puzzle solving with progressive mechanic unlocks, requiring players to manipulate environmental objects, activate pressure plates, and navigate increasingly complex challenges before transitioning into a horror experience featuring flashlight-based survival mechanics and enemy AI. The project emphasized gameplay progression, puzzle sequencing, and player interactions.

- Designed and implemented puzzle progression systems by creating pressure plate mechanics, environmental interactions, and level-based objectives, guiding players through increasingly complex gameplay challenges.
- Developed physics based interaction systems using Unity Rigidbody mechanics for object carrying and dragging, enabling players to construct traversal paths and solve environmental puzzles.
- Integrated Unity Input System, coroutines, ScriptableObjects, and Cinemachine components to support modular interactions, responsive controls, and smooth gameplay transitions.
- Structured level flow and mechanic unlock progression by coordinating puzzle placement and environmental design, improving player onboarding and maintaining gameplay pacing throughout the experience.
- Conducted iterative playtesting sessions to evaluate puzzle clarity, interaction responsiveness, and progression difficulty, refining level layouts and reducing player confusion.
- Debugged interaction and physics instability issues using Unity's debugging tools and Visual Studio Code, improving collision reliability and object behavior during puzzle interactions.
- Helped transition gameplay from puzzle-solving to horror encounters by designing progression triggers and environmental sequencing, creating a shift in gameplay tone and objectives.

Titanic Survivor Prediction, 2024

Developed a machine learning classification system in Python using NumPy and pandas to predict passenger survival outcomes from the RMS Titanic dataset. The project analyzed historical passenger records sourced from the Kaggle Titanic dataset to evaluate relationships between demographic, socio-economic, and environmental factors and their influence on survival probability. Through data preprocessing, feature analysis, and comparative model evaluation, the project assessed the predictive performance of logistic regression, decision tree, and gradient descent algorithms.

- Developed machine learning classification pipelines using Python, NumPy, and pandas to transform raw passenger data into structured datasets suitable for predictive modeling.
- Evaluated logistic regression, decision tree, and gradient descent models through comparative accuracy testing, identifying the decision tree as the most effective predictor with 91.92%.
- Applied feature analysis techniques to passenger demographics, socio-economic status, and cabin data, identifying the factors that most strongly influenced survival outcomes.
- Performed data preprocessing and validation by cleaning datasets and refining input features, improving training consistency and model reliability.
- Conducted model validation and accuracy testing using structured evaluation workflows, ensuring prediction results were reliable and repeatable.
- Analyzed correlations between passenger characteristics and survival outcomes through statistical and machine learning methods, generating interpretable insights from historical data.
- Addressed data quality and feature relevance challenges through iterative model refinement, improving predictive performance and reducing model noise.

WORK EXPERIENCE

IT Intern May 2026 - Present

San Diego Police Department | San Diego, CA

Support technology operations, device lifecycle management, and process automation initiatives for the San Diego Police Department using PowerShell, WPF, SQL, Active Directory, and enterprise asset-management tools.

- Enhanced a PowerShell maintenance application by implementing WPF interface improvements, progress tracking, and user-selection controls, improving usability and reducing technician effort during workstation maintenance.
- Reimaged and validated 50+ Panasonic laptops by testing GPS functionality, wireless communication, and network connectivity, ensuring devices met operational deployment requirements.
- Utilized SQL queries and inventory databases to track technology assets, verify device records, and support equipment lifecycle management.
- Collaborated with teams to perform preventative maintenance and support technology deployments across multiple department locations.
- Documented maintenance procedures and validation workflows to improve consistency, repeatability, and knowledge among technical staff.
- Constructed and organized LAN cable installations to support reliable device connectivity.

Barista August 2025 - Present

Starbucks | San Diego, CA

- Deliver exceptional customer service in a high-volume environment using strong communication, adaptability and problem-solving skills.
- Manage multiple concurrent tasks under intense situations, changing priorities to maintain speed and accuracy during peak hours. Adhere to strict standardized procedures, ensuring consistency in quality and detail.

Teaching Assistant - 3D Game Programming July 2024 - June 2025

San Diego State University, Computer Science Department | San Diego, CA

Served as a Teaching Assistant for a 3D game programming course centered on software engineering principles within interactive system development using the Unity game engine and C# in Visual Studio Code.

- Mentored 48 students per semester through clarification of assignment requirements, creation of examples for each project, and guidance when software issues occurred. Advised on software engineering practices within Unity, supporting development of multiplayer games, AI systems, procedural generation tools, and gameplay architectures.
- Debugged runtime failures including null reference exceptions, collision system issues, and inefficient object instantiation patterns using the Visual Studio debugger and structured code review workflows.
- Guided students in building modular gameplay systems using ScriptableObjects, coroutines, prefabs, Unity Input System, and reusable UI architectures with UI Toolkit and Canvas.
- Developed reusable code templates and instructional examples to improve maintainability, reduce redundant logic, and reinforce scalable software architecture practices.
- Assisted students with version control workflows using GitHub, including repository management, collaboration practices, and debugging merge-related issues.
- Conducted playtesting sessions and technical evaluations focused on gameplay stability, system integration, usability, and adherence to software engineering standards.
- Provided technical instruction on rendering pipelines, shader integration, multiplayer synchronization, and gameplay system optimization across desktop and mobile-connected environments.

Lead Coordinator & Player - Aztec Gaming May 2023 - July 2025

San Diego State University | San Diego, CA

- Lead event and tournament organization for over 100 members of the collegiate Valorant Esports division.
- Member of the Overwatch 2 Esports team, contributing to strategy development and in-game decision making.
- Analyzed team compositions and player feedback to improve match balance and team strategies.

Development Assistant September 2021 - August 2025

KPBS | San Diego, CA

- Organize and support 10+ events annually for over 2,500 major-giving KPBS members.
- Maintain and update member databases with 100% data accuracy.
- Manage financial transactions and reimbursements with a focus on precision and confidentiality.